

Airline Delay Analysis (2003–2022)

Project Overview

Airline delays affect millions of passengers each year and create significant operational challenges for airlines and airports. The goal of this project was to identify the primary causes of delays, evaluate airline and airport performance, examine seasonal patterns, and analyze how delays changed over time.

Using SQL and Tableau, I analyzed over 318,000 records covering 33 airlines and 420 airports across a 20-year period.

Business Questions

1. What delay causes contribute most to industry-wide delays?
 2. Which airlines have the highest delay burden relative to flight volume?
 3. Which airports act as operational bottlenecks?
 4. How do delay patterns vary across months and seasons?
 5. How have airline delays evolved from 2003–2022?
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Dataset

Source: Airline Delay Cause Dataset (Kaggle)

Coverage:

- 318,017 records
- 33 airlines
- 420 airports
- 2003–2022

The dataset includes flight volume, delayed flights, cancellations, diversions, and delay causes such as carrier, weather, NAS, security, and late aircraft delays.

Tools Used

- SQL (SQLite)
 - Tableau Public
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Project Process

Data Cleaning & Validation

- Investigated missing values and incomplete records.
- Checked for duplicate records.
- Validated year and month ranges.
- Created a cleaned analysis table for reporting and visualization.

SQL Analysis

- Performed exploratory data analysis (EDA).
- Calculated airline and airport performance metrics.
- Analyzed delay causes and delay trends.
- Created summary tables for each business question.

Data Visualization

- Exported SQL summary tables to CSV.
 - Built visualizations in Tableau Public.
 - Created an interactive dashboard to communicate findings.
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Key Findings

Delay Causes

Late Aircraft Delay was the largest contributor to total delay minutes (37.9%), followed by Carrier Delay (30.6%) and NAS Delay (26.1%). Together, these three categories accounted for nearly 95% of all delay minutes, while Weather (5.2%) and Security (0.2%) played a minor role.

Airline Performance

Delay rates varied significantly across airlines. Atlantic Southeast Airlines (24.5%), JetBlue Airways (23.7%), and Allegiant Air (22.7%) recorded the highest delay rates, while Hawaiian

Airlines recorded the lowest, highlighting substantial differences in delay performance across carriers.

Airport Bottlenecks

Major hub airports — Newark Liberty, LaGuardia, San Francisco International, Chicago O'Hare, and JFK — consistently combined high traffic volume with elevated average delays, making them critical pressure points in the broader flight network.

Seasonal Trends

Delays followed a clear seasonal pattern. June, July, August, and December consistently experienced the highest delay levels, while September, October, and November recorded the lowest. June recorded the highest average delay per flight, with July and December close behind.

Long-Term Trends

Delay performance fluctuated from 2003–2022, rising through the mid-2000s before stabilizing during the 2010s. Delays dropped sharply in 2020 due to reduced pandemic-era travel, then rebounded in 2021–2022 as demand recovered.

Recommendations

Based on the analysis, airlines and airport operators should:

- **Add schedule buffer to early aircraft rotations.** Since Late Aircraft Delay was the largest single cause (37.9% of delay minutes) and operates through cascading effects across an aircraft's daily legs, airlines should build 10–15 minute buffers into the first 2–3 rotations of high-frequency aircraft to prevent early delays from propagating through the rest of the day.
- **Run root-cause benchmarking against top performers.** Atlantic Southeast, JetBlue, and Allegiant should compare their delay-cause breakdown (carrier vs. weather vs. NAS vs. late aircraft) against Hawaiian Airlines' to identify which specific delay category drives the performance gap, rather than treating "high delays" as a single undifferentiated problem.
- **Review scheduling density at bottleneck airports.** Newark, LaGuardia, SFO, O'Hare, and JFK should reassess departure bank scheduling to reduce clustering during peak congestion windows, since NAS delay (a top-3 cause) is largely a traffic-flow issue rather than an individual-carrier issue.
- **Pre-position staffing and aircraft ahead of seasonal peaks.** Since June, July, August, and December are consistent, predictable delay peaks rather than random spikes, airlines should shift extra ground crew, reserve aircraft, and crew scheduling resources into the 4–6 weeks prior, rather than reacting after delays occur.

Dashboard

The interactive Tableau dashboard combines all five business questions into a single view, allowing users to explore delay causes, airline performance, airport bottlenecks, seasonal patterns, and long-term trends.

Interactive Tableau Dashboard:

[View Interactive Tableau Dashboard](#)

Skills Demonstrated

- SQL Querying
- Data Cleaning
- Data Validation
- Exploratory Data Analysis (EDA)
- Business Analysis
- Root Cause Analysis
- Trend Analysis
- Data Visualization
- Tableau Dashboard Development
- Data Storytelling